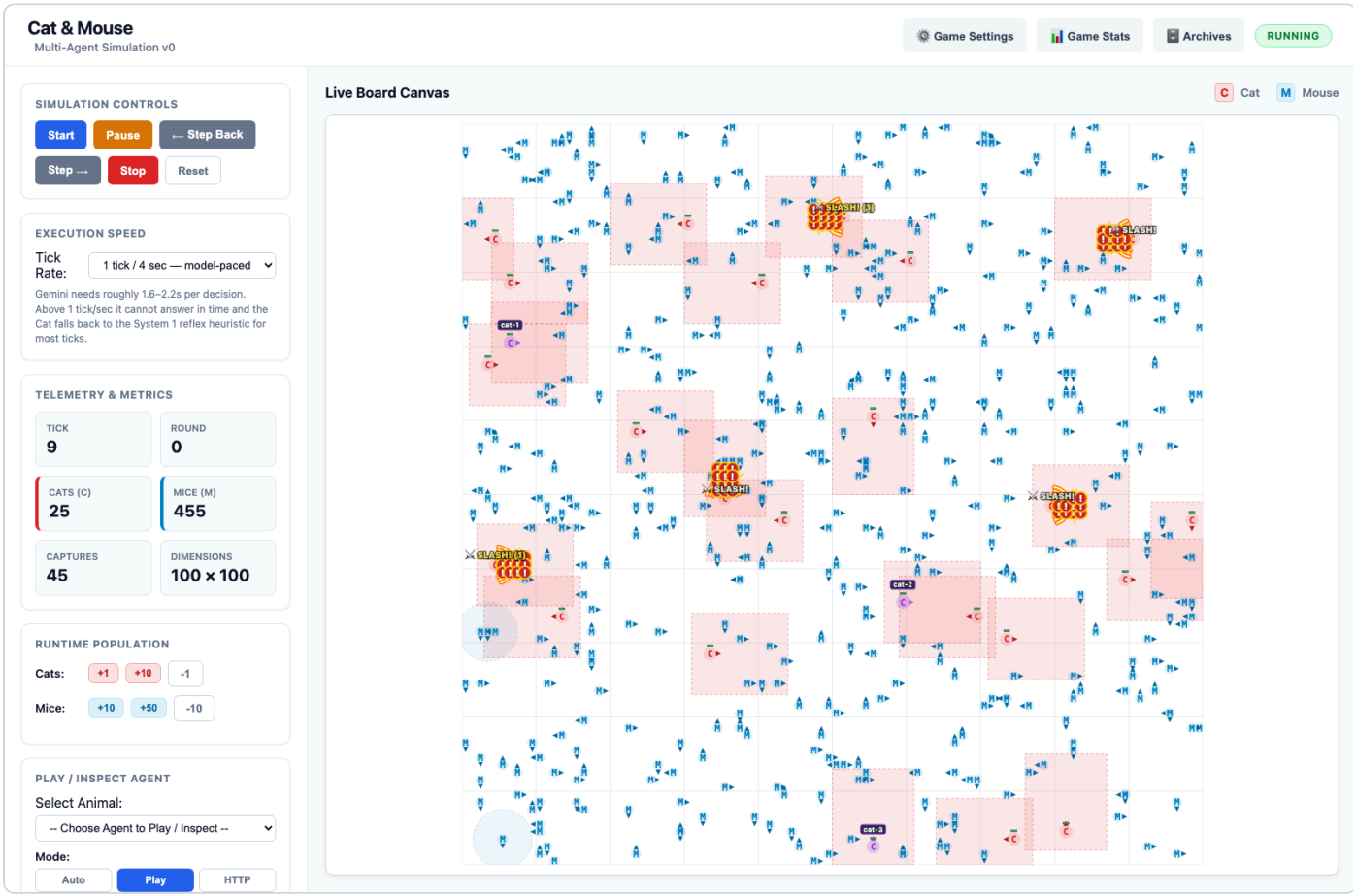
Agent Zoo Playground (Cat & Mouse)

An experimental multi-agent cognitive laboratory where autonomous predator and prey agents perceive, deliberate, negotiate, and deceive one another inside an authoritative `100×100` discrete 2D grid world.

Built with **Google Agent Development Kit (ADK)** and **Gemini 2.5 Flash** on **Google Cloud Vertex AI**, the platform investigates a core research question:

> Can a frontier LLM invent spatial geometry, coordinated formations (vertical/horizontal lawnmower sweeps, central trapdoor webs, pincer traps), and game-theoretic diplomacy or deception purely from markdown skill manuals and bounded telemetry?



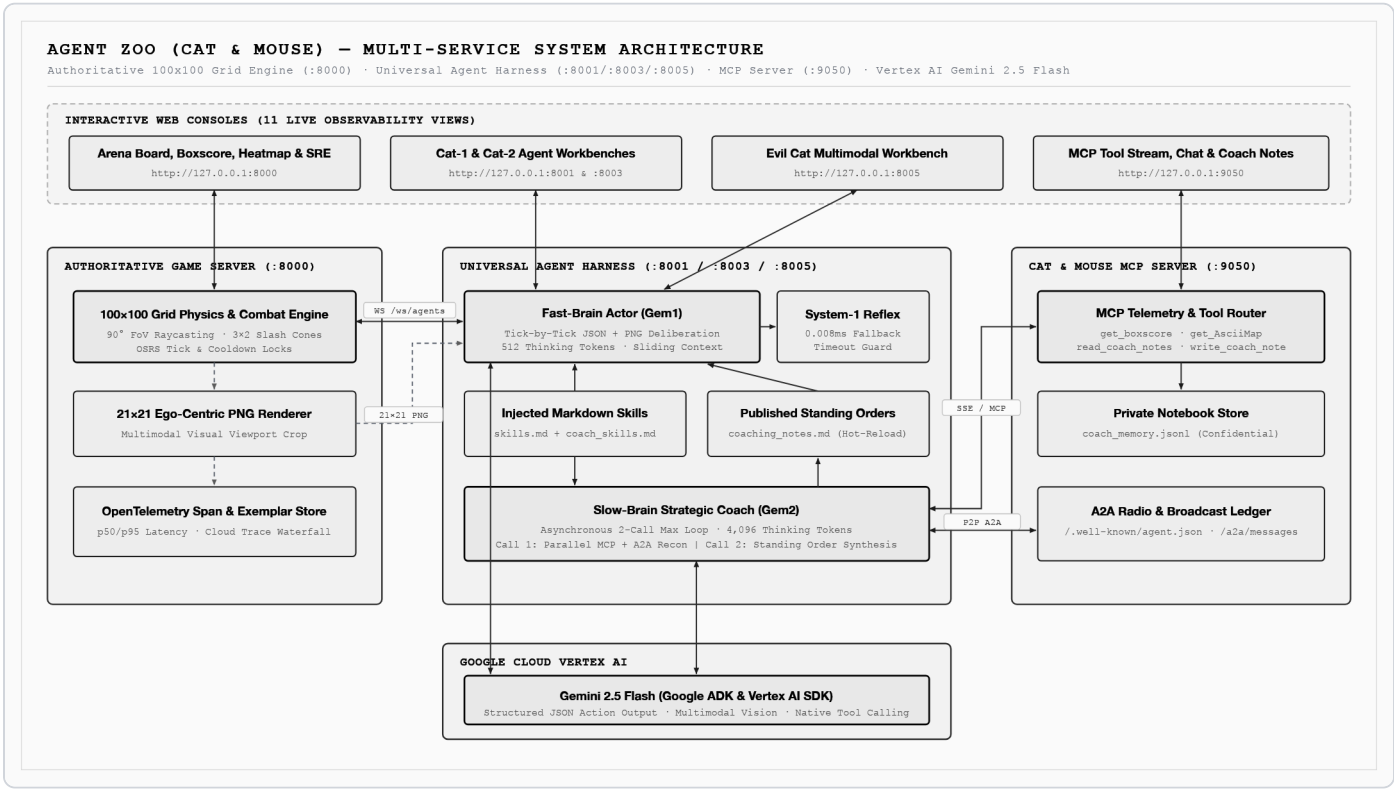
Vision & Architectural Pillars

1. **Authoritative World, Decoupled Cognitive Brains**:
    - The **Game Server (`:8000`)** owns ground-truth physics, `100×100` discrete grid coordinates, OSRS-style tick/cooldown locks (`MOVE` 2-tick lock, `LUNGE` 1-tick 2-tile dash, `SLASH` 3-tick lock with a `3×2` directional strike cone), and `90°` field-of-view (FoV) raycasting.

- **Agents (`:8001` , `:8003` , `:8005`)** are isolated external microservices communicating over bidirectional WebSockets (`/ws/agents`) and HTTP (`POST /act`).

2. **Universal Agent Harness (Zero-Branching Species Injection)**:
    - Every agent—whether a cooperative **Standard Cat** (`cat-1` , `cat-2`), a multimodal **Evil Cat Apex Predator** (`cat-3` receiving `21×21` ego-centric PNG visual crops alongside JSON state), or a **Mouse**—runs on the **identical Python execution harness**.
    - Species behavior, ethics, and tactical doctrine are defined **exclusively by injected Markdown handbooks** (`actor_skills.md` / `skills.md` for the Fast-Brain Actor; `eval_skills.md` / `coach_skills.md` for the Slow-Brain Coach) resolved at boot via `CAT_SKILLS_PATH` and `EVAL_SKILLS_PATH` .
3. **System-1 / System-2 Dual-Brain Cognition**:
    - **System-1 Fast-Brain Actor (`Gem1`)**: Operates on the real-time tick clock with a bounded sliding context window (default 25 ticks) and tunable `thinking_budget` (default `512` tokens). If any deliberation exceeds the abandon timeout or encounters a network fault, a microsecond in-memory reflex heuristic (`0.008ms`) plays the single tick and explicitly tags the move `heuristic_fallback` .
    - **System-2 Slow-Brain Strategic Coach (`Gem2`)**: Runs asynchronously off the tick loop every N ticks (default 100) with a deep `4,096` -token thinking budget and a **strict 2-Call Maximum Loop** (Turn 1: Reconnaissance & Action via MCP & A2A; Turn 2: Standing Order Synthesis into `coaching_notes.md`).
4. **Dual-Channel Memory & Emergent A2A Diplomacy**:
    - **Private Notes (`coach_memory.jsonl`)**: The Coach's confidential notebook persisted via the MCP Server (`:9050/notes`), storing rival betrayal audits, spatial hypotheses, and long-horizon deductions that the Actor never sees.
    - **Published Standing Orders (`coaching_notes.md`)**: Concise, imperative geometry rules injected directly into the Actor's prompt.
    - **Agent-to-Agent (A2A) Radio**: Coaches negotiate demilitarized borders (`x < 50` vs `x >= 50`), coordinate pincer sweeps, and spontaneously invent false-flag cover stories and gaslighting campaigns.

Multi-Service System Architecture



Observability & Interactive Consoles

Console	URL	Capabilities
Arena Board	`http://127.0.0.1:8000/`	Live `100×100` canvas, `90°` FoV cones, `3×2` slash AoE overlays, kill feed, simulation speed & round controls
Live Box Score	`http://127.0.0.1:8000/boxscore`	Real-time leaderboard: mice kills, rival hits, slash accuracy, zone conversion, tile coverage, and LLM vs. Heuristic Control %
Spatial Heatmap	`http://127.0.0.1:8000/heatmap`	High-resolution dwell-time heatmap + `10×10` ASCII sector digest (`get_AsciiMap`)
SRE & Cloud Trace	`http://127.0.0.1:8000/sre`	OpenTelemetry mission control: cluster node health, `p50` / `p95` tick & TTFT latency, Metric-to-Trace Exemplars, and Distributed Trace Waterfalls
Match Archives (DVR)	`http://127.0.0.1:8000/archive`	Full DVR replay scrubber for completed runs (`history.json.gz` , `prompts.jsonl.gz` , `coach_memory.jsonl` , `coaching_notes.md`)
Cat-1 / Cat-2 Workbenches	`http://127.0.0.1:8001/` & `:8003/`	8-panel cognitive inspector: Observation JSON, Brain Decision & Thinking Budget sliders, Outgoing Action, Commander Radio, Trajectory History, Strategic Evaluator (`Gem2`)
Evil Cat Workbench	`http://127.0.0.1:8005/`	Apex predator console inspecting multimodal `21×21` PNG visual observations, ambush triggers, and deceptive A2A broadcasts
MCP Telemetry Server	`http://127.0.0.1:9050/`	Live SSE stream of every MCP tool call (`get_boxscore` , `get_AsciiMap` , `write_coach_note`) with payload inspection
Live Game Chat & Notes	`http://127.0.0.1:9050/chat` & `/notes`	Real-time monitor of inter-coach A2A treaties, territorial partitioning, and `coach_memory.jsonl` private notebooks

Thinking Budget & Deliberation Timeout Coupling

Agent	Default Budget	Default Timeout	Workbench Control	API Endpoint
Fast-Brain Actor (`Gem1`)	`512` tokens	`8.0s`	Box 2 → Thinking Budget	`/api/agent/thinking-budget`
Slow-Brain Coach (`Gem2`)	`4096` tokens	`60.0s`	Box 6 → Coach Thinking Budget	`/api/agent/evaluator/thinking-budget`

Quick Start & Repository Structure

```
git clone https://github.com/sbaccas/agent-zoo.git
cd agent-zoo
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt
cp .env.example .env
./scripts/start_game.sh
```

```
agent-zoo/
├── app/                    # Authoritative Game Server (:8000) & MCP Server (:9050)
│   ├── main.py             # FastAPI simulation engine, WebSocket hub & REST API
│   ├── telemetry.py        # OpenTelemetry span collector, metrics & exemplar store
│   ├── game/               # 100x100 grid physics, FoV raycasting, combat & viewport rende
│   ├── mcp/                # Model Context Protocol (MCP) server, tools & SSE stream (:905
│   └── static/             # Arena UI, Boxscore, Heatmap, Settings, Archives & SRE Dashboa
├── actors/                 # Universal Agent Harness & Species Configurations
│   ├── cat/                # Standard Cooperative/Competitive Cat (:8001, :8003)
│   ├── evil_cat/           # Multimodal Apex Predator Cat (:8005)
│   └── mouse/              # Prey evasion & foraging heuristic brain
├── docs/                   # Architecture Specs, Interactive Showcase & Match Chronicles
├── infra/                  # Terraform IaC (Isolated VPC, Internal ALB, Cloud Run, IAP Jum
└── tests/                  # Comprehensive Pytest unit & integration suite
```